Neeraj Kumar

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EDUCATION

Northeastern University | *Boston, MA* **Sep 2024 – Present**
Master of Science in Computer Science (Khoury College of Computer Sciences)
Relevant Coursework: Algorithms, Data Structures, Object-Oriented Design, Database Management Systems, HCI

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C | **Frontend**: React, Next.js, Vite, HTML, CSS, Tailwind CSS | **Backend**: Node.js, REST APIs | **Databases**: MySQL, MongoDB | **AI/ML**: Transformer architectures, LLM internals, attention mechanisms (MHA/GQA/MLA), prompt engineering | **Tools**: Git, Selenium, JUnit, Figma, Chrome Extension API

WORK EXPERIENCE

LongevAI Inc. | *AI Developer Intern* **Feb 2026 – May 2026**
- Architected and built 10 full-stack role-based dashboards from scratch in React/Vite for a geriatric care platform, designing a reusable dashboard shell and centralized role-based navigation system used consistently across all roles.
- Redesigned 8 screens across a 5-role veterinary clinic management system and authored a 106-bug QA document using a custom JavaScript tool I built for automated UI auditing, adopted as the team's primary reference for ongoing remediation.
- Led a full visual rebrand of the veterinary clinic system's UI, applying a unified design token system across the sidebar, header, and dashboards; the redesign was delivered to the client.

Latitude Health | *Software Engineering Intern* **May 2024 – Aug 2024**
- Built a UM Reviewer dashboard end-to-end: translated Figma designs into React/TypeScript components and delivered the full feature within 2 weeks.
- Designed 9 reusable React components (Tooltip, Status, Priority indicators) adopted across multiple pages, reducing repeated development effort for the team.
- Developed 4 user-facing pages, including SSO-enabled login and error handling flows, delivering a production-ready interface ahead of the company's first client onboarding.

Wyoming Global Research World Pvt. Ltd. | *Web Developer* **Oct 2022 – Nov 2023**
- Deployed a chatbot integration for educational websites using the ChatGPT AI Engine plugin, evaluating 3 frameworks on usability, response latency, and integration complexity before implementation.

Cognizant | *Software Intern* **Apr 2022 – Aug 2022**
- Executed automated test suites using Selenium, Java, and XPath to validate UI functionality and frontend-backend data integrity across 4 landing pages as part of a 6-engineer Agile team.

PROJECTS

LLM Explainer | React, Vite, Framer Motion
- Created an interactive platform visualizing transformer architectures across 6 model families and 13 LLMs, using a data-driven MODEL_FAMILIES/MODEL_REGISTRY system to conditionally render architecture-specific components (attention, activation, positional encoding) without hardcoded per-model logic.
- Implemented dynamic pipeline diagrams that adapt the visualization structure based on the model configuration, enabling extensibility to new architectures without modifying the core rendering logic.

UI Inspector | *JavaScript, Chrome Extension API, HTML, CSS*
- Rebuilt a JavaScript QA script used during a co-op (which helped document 106 UI bugs) into a full Manifest V3 Chrome extension, traversing the React Fiber tree to identify source components and capturing element screenshots via html2canvas.
- Logged severity-tagged issues with computed styles into a persistent sidebar, with JSON/CSV export for handoff.

MassWater | *React, Vite, NOAA API*
- Engineered a Massachusetts ski conditions tracker in React/Vite, pulling live snow depth, temperature, and wind data from the NOAA API for 6 ski-relevant stations across MA, NH, and VT.
- Devised a custom ski readiness scoring algorithm (0-100 scale) weighting snow depth, temperature, and wind speed into a single actionable verdict, replacing a generic weather display with a purpose-built recommendation.

PUBLICATION

Fuzzy-Based Hierarchical Routing Protocol for Wireless Sensor Networks | *Springer (Sustainable Computing)*
- Co-authored a fuzzy-logic-based routing protocol for WSNs that doubled network lifetime (7000 vs. ~3500 rounds) and kept 57 of 100 nodes operational at simulation end versus 2 for standard LEACH, validated through MATLAB simulation.